Y. H. SHANILKA THULSHANI YAPA

BSc. Engineering (Hons)-Computer Engineering (UG), Faculty Of Engineering,University Of Ruhuna

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PROFILE

I am a motivated Computer Engineering undergraduate with a strong interest in software development and emerging technologies. I am confident in working with AI-related concepts and tools, and I bring versatility in adapting to different technologies and problem-solving approaches. I have skills in programming, teamwork, and analytical thinking. I am seeking an internship opportunity to gain hands-on experience and contribute to real-world projects.

TECHNICAL SKILLS

Languages
Python, C#, C, Javascript, Kotlin, Dart, Verilog, SQL, Java

Web & Backend
MERN (MongoDB, Express.js, React.js, Node.js), Firebase

DevOps & Tools
CI/CD, Jenkins, Docker, Terraform, Selenium, Postman, Git, AWS

Design & Mobile
Flutter, Figma, Unity, Matlab, Proteus

ACADEMIC QUALIFICATIONS

Faculty of Engineering, University of Ruhuna 2023 – Present
- CGPA : 3.54

Dhammissara National College 2008 – 2021
- 2018 Ordinary level : 8A's and 1B in English medium
- 2019 Ordinary level (Tamil) : B
- 2021 Advanced level : 2A's and 1B in physical science stream

PROJECTS

ARWays – AR Navigation System (Unity Project-Ongoing) 2025 – Present
- Developed **ARWayz**, an augmented reality indoor/outdoor navigation system using **Unity + AR Foundation** and **Flutter**, focused on delivering precise, real-time directional guidance. Implemented AR arrow overlays with spatial mapping concepts, QR-assisted indoor navigation workflows, and Google Maps-integrated outdoor routing with live location tracking. Built core mobile features including permission handling, navigation UI flows, and cross-platform Android/iOS support. Currently planning enhancements such as higher-accuracy indoor positioning, dynamic rerouting, and offline navigation capabilities.

Book recommendation System (Machine learning - 2 member group project) 2025
- Tech stack includes Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn using google collab. Built a machine learning-based book recommendation system using a Kaggle dataset, including data cleaning, feature engineering, and clustering of books into meaningful reader-focused segments. Implemented and compared K-Means and Agglomerative Clustering with Silhouette Score, Davies-Bouldin Score, and SSE/Elbow analysis, then visualized cluster separation using PCA and distribution plots for interpretability. Developed a cluster-based recommendation workflow for suggesting similar titles and packaged trained artifacts for reuse and deployment integration.

LAN File drop (Group project)
- Developed a LAN-based file transfer system with a C++ core enabling fast peer discovery, file sending/receiving, and transfer tracking across local networks. Implemented socket-based networking with UDP broadcast/listening for device discovery, modular sender/receiver components for reliable file delivery, and persistent transfer logging via CSV history, while using CMake-based build configuration to support structured development and deployment on Windows.

Art supply Management system (Database systems - Group project) 2025
- Developed a hybrid inventory management system leveraging **MariaDB** for structured transactional data and **Neo4j** to optimize complex supply-chain relationship mapping through graph-based queries.

Qaleesi Volunteering Platform (MERN - Individual project) 2025
- Developed a web app using MongoDB, Express, React, Node.js with features like user authentication, profile management, posts (text/images), idea CRUD and messaging.

Software Testing & QA Project 2025
- Developed automated web app tests using TDD, BDD, Selenium, Postman, tracked defects with JIRA, and integrated CI/CD.

FoundIT Lost & Found Mobile App (Group project) 2025

- Engineered a mobile application using Kotlin/Flutter to streamline item management; implemented messaging feature using the lost/found item which improved privacy of the users.

LibriCore Library Management System (Devops Individual Project) 2025

- Architected and deployed the Libricore full-stack application, reducing manual deployment effort through Infrastructure as Code (Terraform) and automation (Ansible), while implementing a Jenkins-based CI pipeline and a GitHub Actions CD pipeline with Kubernetes on AWS EC2 to enable reliable, scalable, and zero-touch deployments.

EstateEase (GUI Module - Individual project) 2024

- Developed a full-stack real estate platform using the MERN stack and SQLite, implementing secure user authentication, an automated loan application workflow, and a persistent cart system, while optimizing data handling and enhancing overall system performance, scalability, and user experienc

EstateEase (Desktop app - Individual project) 2024

- Developed a **WPF desktop application** featuring a robust **CRUD-based real estate management system** integrated with **SQLite** for high-performance local data persistence and streamlined property tracking.

FitEase (UI/UX Design - Figma individual project) 2024

- Designed a cartoon-style exercise app interface in Figma with user-friendly navigation and interactive screens.

CERTIFICATIONS

- Completed the "Python for beginners" by open and distance learning, University of Moratuwa.
- "Python for Everybody" specialization offered by University of Michigan on Coursera.
- Completed the "C# basic" certification on Hacker rank.
- "Excel skills for business" offered by Macquarie University on Coursera.

COMPETITIONS

CodeRally 6.0 organized by IEEE CS Student Branch, IIT 2025

- Competed as a member of Team Nova in the advanced category of the virtual hackathon.

Eminence 5.0 organized by University of Ruhuna 2025

- Placed 8th in a multidisciplinary engineering competition (power systems, ML, network design, innovation pitch) as part of a 5-member team (Team Quadra).

IESL day organized by University of Ruhuna 2024

- Placed 5th in the solid works competition organized by Faculty of Engineering, University of Ruhuna as a part of 2-member team (Team Lightfury)

EXTRA CURRICULAR AND VOLUNTEERING

Rextro Exhibition Team lead and Demonstrator 2025

- Led Finance Management Team, served on Exhibition Committee, and acted as a Demonstrator

Interfaculty Sports Meet (UOR) 2025

- Netball (Champions), Basketball (2nd Runner-up)

Engineering Derby (UOR vs USJ) 2025

- Netball (Champions), Softball Cricket (Champions)

Mehewara Volunteer 2024

- Taught O/L Mathematics to underprivileged students in Puttalam District

Netwizards Tournament (UOR) 2023 – 2024

- overall champions

House Captain (School Inter House Sports Meet), Junior prefect 2021

REFERENCES

Dr. Prabath Weerasinghe (B.Sc., M.Sc., Eng., Ph.D.Eng), *Senior Lecturer*, Faculty of Engineering, University of Ruhuna
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Mr. Thiwanka Silva (B.Sc.Eng.(Hons)(Ruhuna)), *Visiting lecturer & Software Engineer*,
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